

Life from Chemistry

An Abiogenesis Experiment for Autonomous Maintenance

The origin of life is usually posed as a transition from nonliving chemistry to biology, but that formulation hides an experimental difficulty. Prebiotic chemistry has advanced far enough to make many formerly hypothetical pieces of the transition experimentally accessible: plausible building blocks can be synthesized, membranes can self-assemble, reaction networks can sustain nonlinear dynamics, and increasingly elaborate protocell systems can combine several life-like functions. Yet the transition itself remains unresolved. Recent reviews still identify the passage from geochemistry to biology—and the absence of a clear boundary between nonliving and living matter—as an open problem rather than a completed piece of origin-of-life science.¹

Part of the difficulty is that no single familiar biological behavior cleanly marks that boundary. Chemical systems can persist far from equilibrium, autocatalytic networks can amplify their own products, compartments can grow or divide, and replicating populations can display heritable variation without thereby answering whether a self-maintaining biological organization has emerged. Origin-of-life research has therefore increasingly treated the problem as one of **integration** rather than the appearance of a single privileged property. Work on protocell development has framed “minimal functional integration” as an experimental challenge, while research on autonomous functional systems and minimal metabolism asks how chemical processes could begin generating and coordinating the material constraints that regulate their own continuation.²

That literature already occupies much of the conceptual ground needed here. In particular, minimal-metabolism research distinguishes a merely coupled reaction network from a system in which chemical processes generate material structures that

¹ Arif Husain Ansari, “Progress and persistent questions in understanding the origin of life on Earth,” *Discover Life* 56, 35 (2026), DOI: 10.1007/s11084-026-09753-5. The review explicitly describes the transition from nonliving chemistry to biology as unresolved and discusses the difficulty created by the absence of a clear nonliving/living boundary.

² Ben Shirt-Ediss, Sara Murillo-Sánchez, and Kepa Ruiz-Mirazo, “Framing major prebiotic transitions as stages of protocell development: three challenges for origins-of-life research,” *Beilstein Journal of Organic Chemistry* 13 (2017): 1388–1395, DOI: 10.3762/bjoc.13.135; Kepa Ruiz-Mirazo, Carlos Briones, and Andrés de la Escosura, “Chemical roots of biological evolution: the origins of life as a process of development of autonomous functional systems,” *Open Biology* 7 (2017): 170050, DOI: 10.1098/rsob.170050.

act back on and control those processes. It explicitly places such systems at the interface between complex chemistry and biology and notes that an operational criterion for knowing when the relevant autonomous organization has actually been achieved is not obvious.³ The proposal developed here therefore does not claim that self-maintenance, autonomy, or organizational closure is a new solution to abiogenesis. It starts from the opposite premise: these ideas have already brought the field close enough to the boundary that a more discriminating experiment may now be possible.

The remaining difficulty can be seen by separating **resources** from **organization**. Any plausible living or protocellular system is chemically open. It may require externally supplied matter, free energy, solvent, temperature, mineral surfaces, light, or recurring environmental conditions. Dependence on such inputs does not by itself make the system nonautonomous. A different situation arises when the experimenter or environment repeatedly supplies the catalyst, membrane repair, gradient restoration, spatial organization, or other load-bearing control that the candidate system would otherwise have to regenerate for itself. In the first case, the environment feeds the organization. In the second, the environment is performing part of the organization.⁴

This distinction suggests a narrower way to approach the chemistry-to-Life boundary. The boundary cannot be located merely by observing the first appearance of activity, persistence, boundedness, reproduction, or evolvability. It may become experimentally discriminable when a chemical organization begins to generate and regenerate enough of its own load-bearing controls that those controls—not continuing external organizational replacement—carry the organization forward. The immediate question is therefore not simply whether the chemistry continues. It is: **when load-bearing material is lost, diluted, degraded, or replaced, does a bounded chemical organization itself carry forward the controls required for its continued organizational competence?**

That question is stricter than asking whether a system can synthesize more of its own material. New lipid, catalyst, membrane area, or even new compartments may be

³ Nino Lauber, Christoph Flamm, and Kepa Ruiz-Mirazo, “Minimal metabolism?: A key concept to investigate the origins and nature of biological systems,” *BioEssays* 43 (2021): e2100103, DOI: 10.1002/bies.202100103. The paper explicitly distinguishes minimal metabolism from ordinary coupled reaction networks through endogenous production of material constraints and notes that an operational completion criterion is not obvious.

⁴ Ibid.

produced while the organization responsible for that production is repeatedly reintroduced from outside, remains dependent on an unreplaced seed, or disappears and reappears in newly nucleated carriers. Conversely, continued organization does not require the same molecules to remain in place. Living cells routinely replace the material from which they are made. The experimental burden is therefore to distinguish **material persistence** from **organizational continuation** without defining the latter by the very biological status the experiment is meant to test.

Two systems bring the question within reach

The systems used here are not proposed reconstructions of the historical chemistry by which life arose on early Earth. They are experimentally tractable model systems for testing a narrower organizational question at the chemistry-to-Life boundary: whether maintenance-bearing organization can continue as the material that initially carried it turns over. Their value here is discriminatory rather than historical.

A particularly useful starting point is the synthetic membrane system reported by Hardy and colleagues in 2015. Their vesicles contained a membrane-associated oligotriazole ligand, TLTA, whose copper complex catalyzed two coupled reactions: production of additional TLTA ligand and production of triazole phospholipid. The membrane was not merely a container for those reactions. It solubilized the hydrophobic catalyst and relevant reactants, and the synthesis of additional autocatalyst depended on the lipid environment. The result was a reciprocal chemical arrangement in which catalyst production accelerated membrane-lipid production while the membrane environment supported the catalyst chemistry.⁵

Hardy and colleagues then showed that this chemistry could be propagated far beyond a single round of synthesis. After precursors were consumed, they transferred 10% of the vesicle population into fresh precursor solution and repeated the process. Each transfer reduced the concentration of the original POPC membrane material by 90%; after two transfers, the original POPC seed had fallen from 200 μM to 2 μM . Under optimized conditions the authors repeated the serial-transfer procedure 15 times over more than 500 hours while continued triazole phospholipid formation remained

⁵ Michael D. Hardy, Jun Yang, Jangir Selimkhanov, Christian M. Cole, Lev S. Tsimring, and Neal K. Devaraj, “Self-reproducing catalyst drives repeated phospholipid synthesis and membrane growth,” *Proceedings of the National Academy of Sciences* 112, no. 27 (2015): 8187–8192, DOI: 10.1073/pnas.1506704112. The paper reports simultaneous production of additional membrane-associated TLTA catalyst and triazole phospholipid, and experimentally shows that the lipid membrane promotes the autocatalytic chemistry rather than functioning only as a passive compartment.

detectable.⁶ This is unusually valuable for the present problem because it already places material replacement, regeneration, and continued membrane synthesis in the same experimental system.

The same study also recognized that an increasing vesicle population could arise in more than one way. Hardy and colleagues used an encapsulated, membrane-impermeable fluorescent marker to distinguish formation of daughter vesicles that received contents from a pre-existing carrier from the appearance of newly nucleated vesicles that did not. Time-resolved imaging provided evidence for division-like events with content transfer, while separate observations also supported de novo vesicle nucleation adjacent to pre-existing catalytic membranes.⁷ The distinction is crucial: an experiment that merely counts more vesicles after synthesis could otherwise treat lineage propagation and new-carrier formation as though they were the same phenomenon.

These results bring the chemistry-to-Life question remarkably close to an experimentally tractable form, but they do not yet collapse the distinction. The serial-transfer experiments establish long-run synthesis while the original seed material is progressively diluted at the population level. The imaging experiments establish that both content-transferring division and de novo nucleation can occur. What remains unresolved is how those two kinds of evidence join: through deep turnover of the load-bearing material, does the organization continue in one bounded carrier or an explicit daughter lineage, or is long-run propagation instead carried by repeated production of new vesicles? The chemistry persists in either case. The organizational claim is different.

Matsuo and colleagues provide an important counterpoint because their 2019 system is simpler. They constructed a monocomponent synthetic phospholipid liposome whose pre-existing anionic membrane acted as the catalytic surface for production of additional membrane lipid from supplied precursor. No separate catalytic amphiphile had to be embedded in the membrane for that reaction. After precursor addition, the

⁶ Ibid. In the serial-transfer experiments, 10% of the vesicle population was transferred to fresh precursor solution; each transfer reduced original POPC by 90%, two transfers reduced the initial POPC concentration from 200 μ M to 2 μ M, and the procedure was repeated 15 times over more than 500 hours.

⁷ Michael D. Hardy, Jun Yang, Jangir Selimkhanov, Christian M. Cole, Lev S. Tsimring, and Neal K. Devaraj, “Self-reproducing catalyst drives repeated phospholipid synthesis and membrane growth,” *Proceedings of the National Academy of Sciences* 112, no. 27 (2015): 8187–8192, DOI: 10.1073/pnas.1506704112. Hardy et al. used encapsulated HPTS to distinguish content-transferring division from nucleation and time-resolved microscopy to observe division events; separate observations also supported de novo vesicle formation adjacent to pre-existing catalytic vesicles.

liposomes underwent morphological changes including budding followed by division, leading the authors to describe the system as a sustainable self-reproducing liposome.⁸

For the present experiment, Matsuo is not merely a second example of membrane reproduction. It is an adversarial case. The Hardy system suggests an intuitive picture in which distinguishable catalytic and membrane functions become mutually supportive. Matsuo asks whether that differentiation is actually necessary. If a simpler membrane can regenerate the material that performs its own load-bearing catalytic role and carry that organization through turnover, then any proposal that quietly requires several differentiated controls has made the boundary too strong. If the simpler system fails while the Hardy architecture succeeds under the same admission rule using system-appropriate assays, that difference becomes scientifically informative rather than definitional.

The two systems therefore expose complementary sides of the same unresolved question. Hardy supplies a chemically differentiated architecture with unusually strong evidence for repeated self-synthesis, material dilution, vesicle growth, division, and nucleation. Matsuo supplies a simpler architecture capable of testing whether such differentiation is required at all. Neither should be declared living or nonliving in advance. Instead, the experiment must determine what is actually carried forward when the original load-bearing material ceases to be a sufficient explanation of the observed organization.

Following material and carrier separately

The proposed experiment begins by refusing to let material composition and organizational continuity answer the same question. They are related, but they are not interchangeable. A later vesicle may contain mostly newly synthesized membrane while having no ancestral relation to an earlier carrier. Conversely, a continuously tracked vesicle may persist while most of its original load-bearing material is gradually replaced. To distinguish those possibilities, the experiment needs two independent evidence channels: one that determines **where the load-bearing material came from**, and another that determines **what happened to the carrier that bore it**.

⁸ Muneyuki Matsuo, Shota Ohyama, Kotoba Sakurai, Taro Toyota, Kentaro Suzuki, and Tadashi Sugawara, “A sustainable self-reproducing liposome consisting of a synthetic phospholipid,” *Chemistry and Physics of Lipids* 222 (2019): 1–7, DOI: 10.1016/j.chemphyslip.2019.04.007. The pre-existing anionic membrane acted as a catalytic surface for formation of additional membrane lipid from precursor, and microscopy and flow-cytometry observations showed budding followed by division.

The first channel is a source or age measurement. Hardy's chemistry already provides a coarse version of this distinction because the starting POPC membrane and the triazole phospholipid synthesized during the reaction are chemically different species. The original study also quantified both new TLTA ligand and triazole phospholipid using liquid chromatography and mass-spectrometric methods.⁹ That existing analytical separation is enough to ask how much newly generated membrane material appears, but it is not sufficient for every part of the present experiment. Newly synthesized TLTA, for example, is chemically the same catalyst species as TLTA present at the start. If the question is whether the catalyst population itself has turned over, chemical identity must therefore be supplemented by source information.

A practical way to obtain that information is a precursor-labeling strategy. Stable isotopes are routinely used to distinguish newly synthesized lipid from pre-existing lipid and to calculate synthesis or turnover from isotope incorporation measured by mass spectrometry.¹⁰ Applied here, an isotopically distinguishable precursor could be introduced at a defined point in the experiment so that newly formed catalyst or membrane product carries a detectable source mark while the initial material does not. The exact isotope, precursor, labeling fraction, and analytical instrument should be selected around the chemistry rather than prescribed by the contract. The scientific requirement is simpler: the assay must be able to distinguish material inherited from the starting system from material synthesized during the observation period, without materially changing the reactions being measured.

If spatial information becomes necessary, isotope-sensitive vibrational imaging provides a possible higher-resolution extension. Stimulated Raman scattering has been used with deuterated metabolic precursors to visualize newly synthesized lipid in individual living cells with spatial and temporal resolution unavailable from bulk mass

⁹ Michael D. Hardy, Jun Yang, Jangir Selimkhanov, Christian M. Cole, Lev S. Tsimring, and Neal K. Devaraj, "Self-reproducing catalyst drives repeated phospholipid synthesis and membrane growth," *Proceedings of the National Academy of Sciences* 112, no. 27 (2015): 8187–8192, DOI: 10.1073/pnas.1506704112. Hardy et al. used LC-MS/ELSD and related chromatographic measurements to follow synthesis of TLTA ligand and triazole phospholipid; the starting POPC and synthesized triazole phospholipid are chemically distinguishable membrane components.

¹⁰ Satu Hänninen, Pentti Somerharju, and Martin Hermansson, "Metabolic Heavy Isotope Labeling to Study Glycerophospholipid Homeostasis of Cultured Cells," *Bio-protocol* 7, no. 9 (2017): e2268, DOI: 10.21769/BioProtoc.2268. The protocol demonstrates the established use of stable heavy-isotope precursors with mass spectrometry to quantify synthesis, remodeling, and turnover of individual phospholipid species. This is methodological precedent for the source/age logic proposed here, not a claim that its specific labeling chemistry transfers directly to Hardy or Matsuo without validation.

spectrometry.¹¹ That does not make Raman imaging a requirement for this experiment. A matched chemical assay may be preferable if it perturbs the Hardy or Matsuo systems less. The important point is that source resolution is technically separable from ordinary fluorescence labeling: one assay asks **which material is new**, while another can ask **which carrier it belongs to**.

That second channel is ancestry. Here the experiment follows the bounded structure rather than the molecular source. A tracked carrier might remain continuously observable while its membrane composition changes. It might bud or divide and transfer the relevant organization into a daughter. It might disappear while an unrelated vesicle nucleates nearby. The bounded structure might dissolve and later reform, or the population as a whole might continue even though the ancestry of individual carriers cannot be resolved. Those outcomes cannot be collapsed into a single category called “self-reproduction,” because each supports a different continuity claim.

Hardy’s original microscopy already demonstrates why this distinction is experimentally realistic. Content labeling and time-resolved imaging were sufficient to observe division-like events in which encapsulated material passed into daughter structures, while other observations supported de novo vesicle formation.¹² The proposed experiment extends that discrimination across the turnover interval. The ancestry channel must follow a carrier, or an explicit parent-to-daughter lineage, long enough to determine whether the organization observed after substantial material replacement belongs to that tracked history or instead to newly formed carriers.

The two channels do not have to be measured by the same instrument, and a credible first implementation need not demand single-vesicle mass spectrometry. Parallel matched arms are scientifically acceptable if the experimental design preserves the same chemistry and supports a defensible linkage between the source-resolved population and the carriers whose ancestry is being observed. One arm could quantify the changing old-to-new material fraction with chromatography or mass spectrometry

¹¹ Junjie Li and Ji-Xin Cheng, “Direct visualization of de novo lipogenesis in single living cells,” *Scientific Reports* 4 (2014): 6807, DOI: 10.1038/srep06807. The study combined deuterium-labeled precursor chemistry with stimulated Raman scattering microscopy to spatially resolve newly synthesized lipid in individual cells. It is cited here as proof of measurement principle for source-sensitive live imaging, not as a prescribed protocell assay.

¹² Michael D. Hardy, Jun Yang, Jangir Selimkhanov, Christian M. Cole, Lev S. Tsimring, and Neal K. Devaraj, “Self-reproducing catalyst drives repeated phospholipid synthesis and membrane growth,” *Proceedings of the National Academy of Sciences* 112, no. 27 (2015): 8187–8192, DOI: 10.1073/pnas.1506704112.

while a matched imaging arm resolves carrier histories. A carrier-level conclusion, however, requires enough cohort linkage to exclude an untracked subpopulation of newly nucleated carriers as the source of the bulk turnover signal; if that exclusion cannot be made, the result remains population-level or indeterminate. A stronger later implementation could combine source-sensitive imaging and lineage tracking in the same carriers. What cannot be done is to infer ancestry from bulk turnover, or to infer deep material turnover merely because a vesicle lineage remains visible.

Even source resolution and ancestry together are not enough. The experiment must also defeat a **passive-seed alternative**. Suppose a tracked vesicle continues to function while 90%, 99%, or even more of some initial material has disappeared. None of those percentages is intrinsically decisive. A small residual population of old catalyst could still perform the load-bearing chemistry, or a small fraction of original membrane could remain the critical organizing scaffold. Declaring an arbitrary replacement threshold would therefore convert a measurement convenience into a biological criterion.

Instead, the threshold should be functional. The old material must become insufficient to explain the observed maintenance relation. That could be shown by independently determining the concentration below which the original catalyst can no longer sustain the relevant synthesis rate, by showing that turnover kinetics leave too little old material to account for the observed function, or by perturbing the newly synthesized material and demonstrating that the maintenance relation now depends on it. The exact route can differ between systems, but the logic cannot: **new material counts as organizational replacement only when the retained old material is no longer a sufficient explanation of the function being attributed to the continuing system.**

This point is especially important in Hardy. The published serial transfers already drive the original POPC concentration down dramatically at the population level, which establishes a powerful material-dilution result.¹³ But population-level dilution does not by itself show that one carrier or daughter lineage remained organizationally continuous through that replacement, and it does not by itself establish that the regenerated material rather than a residual seed carried the decisive function. The proposed experiment joins those previously separate questions. Matsuo creates the

¹³ Ibid.

complementary challenge: because newly produced membrane is of the same synthetic phospholipid species as the starting membrane, source labeling becomes even more important if old and newly synthesized membrane are to be distinguished during an otherwise compositionally continuous lineage.

Once those three burdens are met—source resolution, carrier ancestry, and passive-seed exclusion—the experiment can ask the question that matters most: whether the **newly generated material actually takes over the load-bearing maintenance function**. That final step turns a turnover measurement into an organizational test.

Showing that new material appears in an ancestral carrier is still not enough. The regenerated material must participate in the relation that makes continued organization possible. In Hardy, for example, newly synthesized phospholipid may enlarge a membrane without being the material that sustains the catalytic environment, while newly synthesized TLTA may accumulate without becoming functionally important at the relevant concentration or location. The decisive evidence is therefore causal rather than compositional: when the original load-bearing material has become insufficient, does the newly generated material now perform the function on which continued synthesis and organization depend?

That burden can be approached through matched perturbations rather than by assigning biological significance to turnover itself. A control in which new synthesis is blocked while the initial carrier and environmental resources remain available establishes how long the starting organization can persist without regeneration. A complementary intervention can selectively reduce or interrupt the contribution of newly generated material after substantial turnover has occurred. If continued function then fails while matched unperturbed carriers continue, the new material has stronger standing as a functional replacement rather than a by-product of an otherwise seed-driven process. The intervention need not destroy the entire vesicle. Its purpose is to discriminate what is carrying the maintenance relation at the time the claim is made.

The same logic requires a control for new-carrier formation. Conditions that permit de novo nucleation should be characterized independently enough that newly appearing vesicles can be recognized rather than folded into an apparent lineage. A population may remain chemically productive for many generations while individual

carriers repeatedly disappear and are replaced by newly nucleated structures. That result could be important in its own right, especially for models of prebiotic propagation, but it is weaker evidence for continued organization within a bounded carrier or inherited lineage. The experiment should preserve that weaker result rather than promoting it simply because the population survives.

The measurements themselves also become part of the scientific burden. Stable-isotope labeling, fluorescence markers, confinement, surface immobilization, microfluidic trapping, repeated illumination, or other techniques used to observe the system may alter membrane behavior or reaction kinetics. A label that changes catalyst affinity, a trap that prevents normal division, or an imaging geometry that suppresses nucleation would make the system easier to measure by changing the thing being measured. Labeled and unlabeled chemistry should therefore be compared for the relevant reaction and membrane behaviors, and tracked or confined carriers should be compared with otherwise equivalent unconfined populations whenever the observation method could plausibly affect the organization.

This is more than an ordinary technical control. Experiments can discriminate between candidate organizations only to the extent that the chosen observations and interventions are capable of separating them. Recent formal work on experimental probes of autopoietic self-maintenance makes the same distinction from another direction: a causal organization and the experimental response profile used to investigate it are not identical, and an assay panel warrants organizational discrimination only relative to the alternatives it is actually capable of distinguishing.¹⁴ The present proposal adopts that experimental discipline without requiring the accompanying formalism. An ambiguous assay returns an ambiguous result; it does not become evidence for absence.

Taken together, these requirements define what will be called the **Organizational Continuity Through Turnover Test**, or OCTT. OCTT asks whether a candidate system can be followed across replacement of the material on which its organization depends while independently resolving the source of that material, the ancestry of the carrier, the insufficiency of the original seed, and the functional contribution of what

¹⁴ Juan J. Segura, “Experimental probes of autopoietic self-maintenance,” *BioSystems* 268 (2026): 105928, DOI: 10.1016/j.biosystems.2026.105928. Segura explicitly separates causal organization from the experimentally observed response profile and argues that restricted experimental probes discriminate organizational models only when the available protocol family is adequate for the candidate alternatives. This is used here as support for assay adequacy and model-relative experimental discrimination, not as the source of OCTT or AMC.

has been regenerated. It does not ask whether every molecule has been replaced, and it assigns no privileged percentage of turnover. The relevant horizon is reached when the old material has ceased to be a sufficient explanation of the function under investigation.

OCTT can therefore return several scientifically distinct results. One bounded carrier may remain continuously operative through the turnover interval. A parent may instead divide and pass the relevant organization into a daughter lineage. A new carrier may nucleate without the required ancestry relation, the original carrier may dissolve before a later organization appears, or the population may continue while individual ancestry remains unresolved. If the evidence cannot distinguish among those possibilities, continuity remains indeterminate. None of these outcomes needs to be redescribed to protect the original hypothesis.

Even the strongest OCTT result, however, is not yet a claim that autonomous maintenance has been demonstrated. A vesicle could replace its membrane while some essential control remains externally supplied. It could carry one regenerated function while the rest of the organization depends on experimental intervention. Or a causal loop could be present without being sufficient to preserve the organization across the declared conditions and observation horizon. OCTT establishes what kind of organizational continuity has survived material turnover. A second question must then ask **what that continuing organization is actually maintaining**.

For that purpose, the experiment uses a separate admission test called the **Autonomous Maintenance Core**, or AMC. An AMC claim requires more than synthesis and more than continuity. The candidate controls must demonstrably affect the processes on which the organization depends; the system must materially contribute to producing or regenerating those controls rather than receiving their organizational function from outside; those relations must be integrated strongly enough to support one another at the declared level; regeneration must matter against loss or dilution rather than merely coexist with passive persistence; and the organization must retain the capacity to continue those relations across the declared environment and observation horizon. The measurement family must also be capable of distinguishing that organization from the relevant alternatives.

Separating OCTT from AMC prevents two opposite errors. Material turnover cannot be promoted into autonomous maintenance merely because a carrier persists, and an

apparently self-maintaining chemical network cannot claim continuity through turnover when the evidence tracks only population averages or continuing synthesis. OCTT asks **what continues**. AMC asks **whether what continues carries enough of its own maintenance organization to remain competent as that organization**. Only when those two questions are kept separate can the experiment approach the chemistry-to-Life boundary without placing the desired answer inside the definition of the assay.

What the outcomes would mean

The value of the experiment depends on deciding what each result means before the result is known. Otherwise, a system that fails one version of the hypothesis can always be redescribed after the fact as having demonstrated some neighboring form of continuity or self-maintenance. The proposed interpretation is therefore deliberately conservative: each result is allowed to support only the strongest claim actually warranted by the measurements.

The strongest continuity result would be a carrier that remains continuously tracked while the original load-bearing material falls below its functional sufficiency threshold and newly generated material assumes the relevant maintenance role. That result would establish continuous bounded carry through turnover. It would show that organizational continuation need not depend on persistence of the original material. It would not, by itself, establish autonomous maintenance, because the continuing carrier might still depend on externally supplied organizational functions that the experiment has not yet excluded.

A second strong result would occur if the original carrier divides and an explicitly tracked daughter inherits the maintenance organization through turnover. That would support lineage continuity rather than uninterrupted persistence of one carrier. The distinction matters. A daughter need not be treated as literally identical to its parent for the experiment to establish that the organization has been carried forward through an ancestry relation. The scientific claim is continuity of an organized lineage, not metaphysical sameness.

A different result would occur if newly synthesized material supports the appearance of new vesicles without the required ancestry relation to the previous carrier. That is *de novo* carrier production. It may be highly relevant to abiogenesis, because a

chemistry capable of repeatedly generating organized compartments could provide a powerful mode of prebiotic propagation. But it does not establish that one bounded organization, or one inherited lineage, maintained itself through turnover. The same caution applies if the original carrier disappears and a later organization reconstitutes. Reappearance of a similar organization is not automatically continuity of the prior one.

The weakest positive population-level result would be sustained propagation in which the ensemble continues to generate functional carriers but individual ancestry cannot be resolved. Such a result should remain a population-level claim. It could reveal a robust chemical ecology in which organized structures continually arise and sustain the reaction network collectively. What it cannot establish is carrier-level continuity simply because the population never goes to zero. If source, ancestry, passive-seed, or functional evidence remains unresolved, the correct result is indeterminate rather than negative.

These distinctions become more consequential when the continuity result is passed to the autonomous-maintenance question. A strong OCTT result says that an organization has survived material turnover in a specified way. The Autonomous Maintenance Core asks whether that continuing organization also carries the controls required for its own continued competence. For that stronger admission, the candidate control must affect the relevant processes, the system must contribute materially to producing or regenerating that control, the maintenance relations must be integrated at the declared level, regeneration must matter against actual loss or dilution, and the organization must remain capable of continuing those relations over the declared horizon. A positive continuity result can therefore coexist with an indeterminate AMC result.

Hardy provides the clearest route to a positive differentiated-maintenance result because the candidate organization contains distinguishable catalytic and membrane roles that already support one another chemically. If the proposed experiment shows continuous carrier or daughter-lineage continuity through deep source-resolved turnover, defeats the passive-seed alternative, demonstrates that regenerated catalyst and membrane material assume the relevant functions, and closes the remaining autonomous-maintenance burdens, then the system would satisfy the proposed AMC admission at that declared scope. That conclusion would be stronger than saying that

the vesicles grow, divide, or synthesize their own components. It would say that the continuing organization materially carries the regeneration of the controls required for its own continued organizational competence.

Matsuo is designed to make that interpretation vulnerable. Its simpler candidate architecture asks whether a membrane-associated catalytic role can carry enough of the same maintenance burden without the differentiated catalytic arrangement seen in Hardy. If Matsuo achieves comparable source-resolved continuity and independently closes the same maintenance burdens, then the hypothesis that autonomous maintenance requires several differentiated controls cannot be retained merely because Hardy happens to contain them. The simpler system must be admitted under the same rule. Conversely, if Hardy closes the maintenance contract while Matsuo repeatedly fails at a specific burden under comparable evidence, the difference identifies an experimentally meaningful candidate requirement rather than a requirement inserted into the definition beforehand.

This is why the Matsuo arm is not a secondary replication experiment. It is an adversarial test of the boundary itself. A useful admission rule must remain open to the possibility that the minimal organization is simpler than expected. If the one-control candidate architecture is sufficient, the experiment should reveal that. If differentiated controls are required, that conclusion should emerge from failure of simpler systems under the same admission rule using system-appropriate assays.

The experiment must also remain vulnerable to a more damaging result: a system that is unambiguously nonliving under the relevant scientific comparison may satisfy the same AMC contract. If that occurs, the correct response is not to redefine the comparator as living. It would show either that autonomous maintenance is broader than Life or that the proposed maintenance contract is too weak to perform the intended discrimination. In either case, the AMC-to-Life step would have to remain open.

That final separation is essential. Even a fully positive AMC result would not automatically establish Life. Autonomous maintenance is the domain object being tested; Life is the larger claim. The experiment therefore permits another lawful outcome in which a system satisfies the maintenance contract but fails the later Life translation because the relevant boundary, level, continuation capacity, or failure discrimination does not carry across. Such a result would not erase the

autonomous-maintenance finding. It would show that the chemistry-to-Life boundary lies beyond it.

The result structure is therefore intentionally asymmetric. Strong claims require the conjunction of several independent evidentiary burdens, while weaker outcomes are allowed to remain weaker. Continuous carrier turnover does not become autonomous maintenance by itself. Autonomous maintenance does not become Life by itself. Population propagation does not become lineage continuity. De novo nucleation does not become persistence of the prior carrier. And inadequate measurement does not become evidence of absence. The experiment is designed to preserve those distinctions whichever way the chemistry answers.

Closing the contract before running the experiment

The experiment is intended to remain open scientifically while being closed procedurally. Those two forms of closure are different. Scientific closure would require the chemistry to be tested and the evidence to decide which outcome has actually occurred. Procedural closure means that the evidentiary burdens, permissible result classes, and consequences of those results are specified before the experiment supplies an answer. The present work claims the second and not the first.

That distinction matters because the hardest cases lie near category boundaries. A productive protocell system can look increasingly life-like as more functions accumulate, and the temptation to reinterpret an ambiguous result grows with the investment in the hypothesis. Predeclaring the result grammar limits that flexibility. If source or ancestry cannot be resolved, the result remains indeterminate. If propagation occurs only through new-carrier formation, it remains new-carrier propagation. If organizational continuity survives turnover but one or more autonomous-maintenance burdens remain open, the system does not receive an AMC admission simply because the continuity result is impressive. If AMC is eventually admitted but the later Life translation fails, the autonomous-maintenance result remains standing while the Life claim remains open.

The contract has been encoded in Lean for precisely this reason. The Lean contract/adaptor does not model the chemistry itself and does not predict the result of the proposed experiment. Instead, it formalizes the logical relations among the evidentiary states. It contains the OCTT result classes, the AMC admission conditions,

the current Hardy and Matsuo literature states, the prospective experimental branches, and the separate translation from an admitted autonomous-maintenance object to the scoped Structural Flow Life object. The human-readable companion exposes and maps that formal surface for audit and interpretation. The purpose is not to make Lean an arbiter of biology. It is to make it harder for the interpretation rules to move unnoticed after the evidence arrives.

The current machine state preserves the empirical openness of the project. The published Hardy evidence remains indeterminate under the full OCTT and AMC contracts, as does the published Matsuo evidence. At the same time, the machine contains prospective result branches in which a future Hardy or Matsuo experiment discharges the required burdens. Those successful branches can reach a result labeled STRUCTURAL CONFORMS only when the continuity evidence, autonomous-maintenance evidence, and separate translation burdens have all been independently discharged. Encoding such a possible future result does not assert that it has occurred. The same machine package simultaneously preserves the current state as NOT YET ADJUDICABLE.

This separation between a prospective successful branch and the current evidence state lets the contract specify in advance what would count as success without turning that criterion into evidence that success already exists. It also preserves the strong Matsuo branch: if the simpler candidate architecture independently satisfies the same AMC burdens, the machine admits it rather than protecting a differentiated-control hypothesis.

Only after an AMC admission does Structural Flow enter as a translation problem. The scoped Life object used here is not chemical activity, replication, growth, or reproduction. The biological question is whether, at the declared level and scope, the continuing organization itself carries the maintenance relations on which its continued viability depends. The machine contract names that job maintenance-bearing continuation. The proposed translation therefore asks whether an empirically admitted autonomous-maintenance organization preserves that job, whether the relevant boundary and carrier have been identified at the appropriate level, and whether failure of the proposed Life organization can be distinguished from mere loss of chemical activity.

That translation is intentionally capable of failing. A chemistry may satisfy the AMC contract while proving too weak, too broadly distributed, or incorrectly scoped to realize the Life object. A candidate may preserve impressive chemical organization without sustaining the relevant continuation capacity. A population-level organization may prove to be the correct carrier rather than an individual vesicle, or no presently tested level may clear the translation at all. Those are not loopholes. They are possible empirical outcomes. Structural Flow contributes a way to keep the domain result and the larger Life claim from being collapsed into one another before the evidence licenses the connection.

The same discipline applies to identity. Neither continuous turnover nor parent-to-daughter inheritance is treated as proof that an object remains strictly identical through change. The experiment asks narrower questions that chemistry and imaging can answer: whether one bounded carrier remains continuously operative, whether a daughter inherits the relevant organization, whether a new carrier forms without that ancestry, whether the prior carrier dissolves before reconstitution, or whether only the population can be shown to continue. Those continuity relations are scientifically useful without requiring the experiment to settle a metaphysical question about sameness.

The public machine package reflects this limited job. The Life / Origin-of-Life Lean contract and adapter is an append-only companion to the compatible Structural Flow Universal Kernel rather than a modification of the kernel itself. The kernel required no repair for the Life object, the OCTT contract, the AMC contract, the Hardy and Matsuo prospective branches, the C4 dissolution/reconstitution completion, or the translation architecture. The Lean contract/adapter append has been machine-checked clean twice against that unchanged kernel. The public dependency is therefore expressed as the compatible
SF_Life_Origin_of_Life_Contract_and_Adapter_v0_5+ release line rather than as a requirement that every later editorial iteration trigger a chain of file updates.

Machine cleanliness should not be mistaken for experimental validation. Lean establishes that the encoded distinctions and conditional result branches are internally admissible within the declared formal architecture. It cannot establish that a vesicle maintained itself, that a catalyst actually turned over as predicted, that an assay was

experimentally adequate, or that a candidate system is alive. Those are empirical questions and remain empirical questions after formalization.

The proposed experiment is therefore narrower than a declaration that the chemistry-to-Life boundary has been found. Any step may fail: Hardy may prove to propagate primarily through new carriers rather than through continuing carriers or lineages; residual seed material may remain sufficient to explain the chemistry; Matsuo may show that the simpler candidate architecture is enough; a clearly nonliving comparison system may satisfy AMC, showing that autonomous maintenance is broader than Life; or an AMC-positive system may fail the Structural Flow translation. Each result would narrow the question differently, and none requires being promoted into a stronger conclusion than the evidence supports.

The endpoint of the proposal is not “this chemistry is Life.” It is a precommitted experimental question: when the material that originally carried an organized chemistry no longer suffices to do so, what continues, what now carries the maintenance function, and is that continuing organization itself responsible for maintaining the conditions of its continued competence? If the experiment can answer those questions cleanly, the chemistry-to-Life boundary becomes narrower than it was before the experiment, regardless of which branch the evidence selects.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation and revision of this manuscript and its accompanying formal materials, the author used generative AI tools to assist with drafting support, language refinement, structural editing, experimental-design organization, formalization support, machine-audit organization, consistency checking, and clarity. All substantive arguments, scientific claims, experimental-design choices, formalization choices, scope decisions, and final editorial judgments are the author's own. The author reviewed and approved the final manuscript and accompanying formal materials and takes full responsibility for their content.

Life from Chemistry

An Abiogenesis Experiment for Autonomous Maintenance

Signed,

Rick